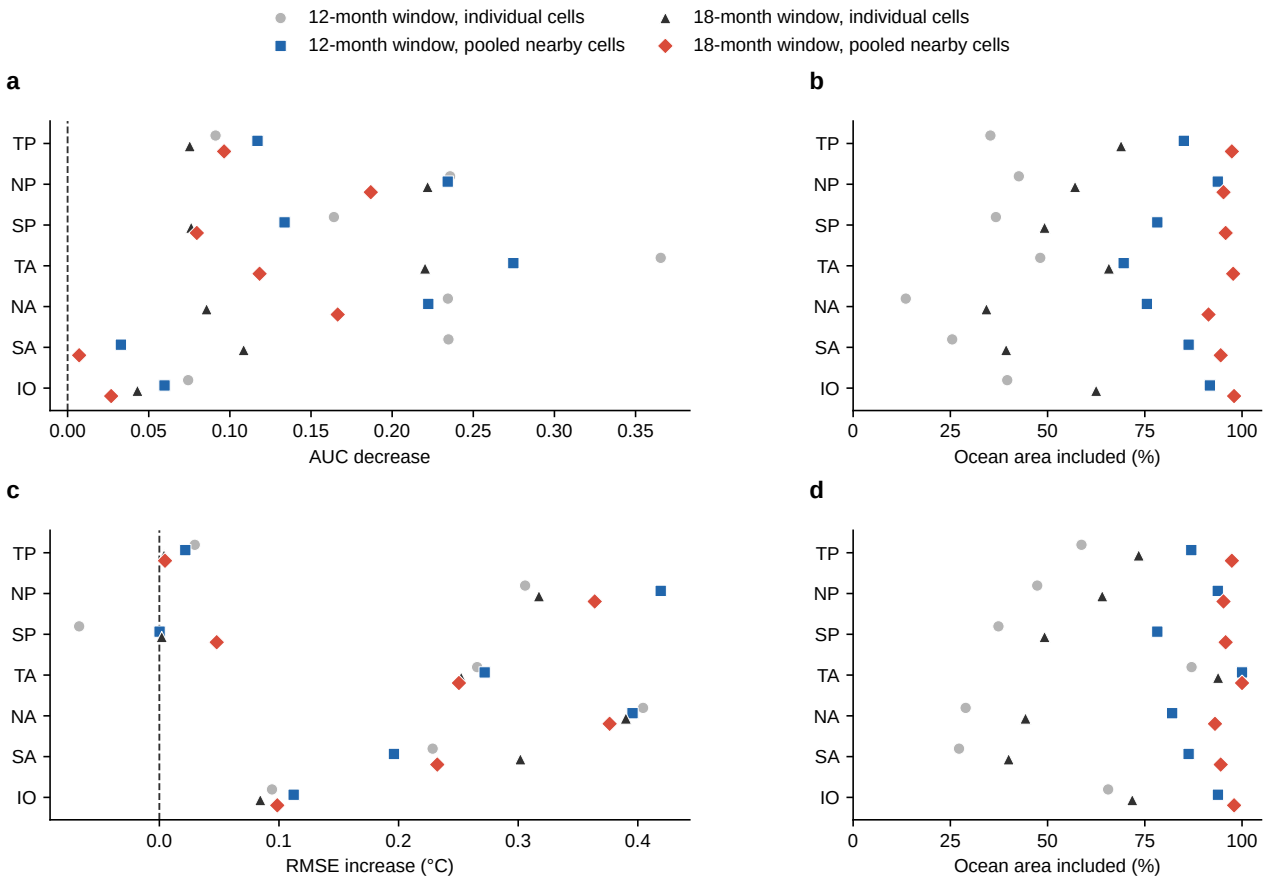
Supplementary Fig. 17 | Sensitivity to verification duration and spatial pooling



Four combinations compare the original July–June window (12 months) and its symmetric three-month extension (April–September of the following year; 18 months), evaluated at individual grid cells or within fixed 11° × 11° neighbourhoods. a,c, Regional AUC decreases and conditional intensity RMSE increases. b,d, Their common valid ocean-area coverage. Red diamonds identify the 18-month, 11° configuration used in Supplementary Figs. 1c and 16. All three events use the same window, lead range and neighbourhood definition. The 12-month, single-cell calculation provides the unpooled reference. All seven fixed regions are retained; no region is removed because its sign differs. Individual-cell estimates use each grid cell separately; pooled estimates combine samples within an 11° × 11° same-basin neighbourhood.

Each point is a region-level summary, not an independent replicate or significance estimate. The longer window changes the seasonal/event-life-cycle support. The larger neighbourhood changes spatial support; higher coverage is not evidence of higher independent spatial resolution or a more accurate forecast. This sensitivity analysis retains the main archived input profile and its documented source-selection limitations.